

Syllabus

Intensive 1: April 26 - May 21

<https://ioai-official.org/wp-content/uploads/2025/10/Syllabus.pdf>

Week 1: 26-30 April (5 days)

Foundations + Supervised Learning + First Neural Network

	level 1	level 2
1	• Python crash course (loops, functions) • NumPy basics • Pandas (loading datasets) Tasks&Materials: TODO	
2	• Matplotlib / Seaborn • Train-test split • Scaling (standardization) Tasks&Materials: TODO	
3	• Regression vs classification • Loss functions Tasks&Materials: TODO	
4	• Decision Trees • Overfitting vs underfitting • Cross-validation Tasks&Materials: TODO	
5	• Perceptron • Gradient descent intuition • PyTorch basics Tasks&Materials: TODO	

Week 2: 02-07 May (6 days)

Computer Vision + CNNs

	level 1	level 2
1	• Pixels, tensors • Data augmentation Tasks&Materials: TODO	
2	• Filters • Feature maps Tasks&Materials: TODO	
3	• Conv → ReLU → Pooling • Build CNN Tasks&Materials: TODO	
4	• Regularization (dropout) • Batch normalization Tasks&Materials: TODO	
5	• Pretrained models like ResNet Tasks&Materials: TODO	
6	• Object detection (YOLO intuition) • Segmentation (U-Net idea) Tasks&Materials: TODO	

Week 3: 09-14 May (6 days)

Sequences → Attention → NLP

	level 1	level 2
1	• Tokenization • Vocabulary	

	Tasks&Materials: TODO	
2	• Word vectors • Meaning in vector space Tasks&Materials: TODO	
3	• Why RNNs fail (vanishing gradients conceptually) Tasks&Materials: TODO	
4	• Attention Tasks&Materials: TODO	
5	• Transformers Tasks&Materials: TODO	
6	• Practice Tasks&Materials: TODO	

Week 4: 16-21 May (6 days)

Unsupervised + Modern AI (CLIP, Self-Supervised)

	level 1	level 2
1	• Clustering • K-Means • Distance intuition Tasks&Materials: TODO	
2	• Dimensionality Reduction • PCA intuition • Visualization Tasks&Materials: TODO	
3	• Self-Supervised Learning	

	• Contrastive learning idea Tasks&Materials: TODO	
4	• CLIP Tasks&Materials: TODO	
5	• GANs • Diffusion models Tasks&Materials: TODO	
6	• Practice Tasks&Materials: TODO	